

LEAN FORMALIZATION OF THE IUT-*abc* PROOF CANDIDATE VERIFIED CORRIDOR, EXACT OPEN INTERFACES, AND THE REQUIREMENTS FOR A GENUINELY CLOSED PROOF

CHATGPT

ABSTRACT. This paper specifies and formalizes in Lean 4 the complete logical dependency corridor of an IUT-based proof candidate for the *abc* conjecture. It defines a standard logarithmic formulation of *abc*, proves the extraction of a coherent pilot witness from equality of composite poly-morphisms, derives the IUT III coefficient estimate $C_\Theta \geq -1$, and connects that estimate to an explicit IUT IV reduction interface whose output is the standard *abc* statement.

The resulting Lean theorem has the form

$$\text{PolyCompatiblePilotData} + \text{IUTIVReduction} \implies \text{ABCConjecture}.$$

The theorem contains no proof placeholders or newly declared axioms. It is nevertheless conditional: the concrete IUT III source instantiation and the complete IUT IV arithmetic-geometric reduction remain theorem arguments rather than closed constructions. This distinction is not cosmetic. A proof assistant checks the theorem actually stated, and cannot transform an assumed structure into a proved instance.

Accordingly, the present development is a rigorous formalization of the verified core and an exact blueprint for a full proof, but it is not yet an unconditional machine-checked proof of the *abc* conjecture. The two remaining interfaces are listed at definition-level granularity.

Central conclusion

A Lean build of the present project verifies every implication after the two explicit interfaces have been supplied. It does not verify that the full IUT definitions construct those interfaces. Therefore it would be mathematically incorrect to title the current result “A fully Lean-verified proof of the *abc* conjecture.”

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Date: August 19, 2026.

2020 Mathematics Subject Classification. Primary 11D75, 11G50; Secondary 03B35, 18A40.

Key words and phrases. ABC conjecture, inter-universal Teichmüller theory, Lean, formal verification, proof interface.

1. THE EXACT abc STATEMENT

For a natural number n , define its radical by

$$\text{rad}(n) = \prod_{p|n, p \text{ prime}} p.$$

For positive pairwise coprime natural numbers a, b, c with $a + b = c$, the logarithmic form of the abc conjecture is

$$\log \max\{a, b, c\} \leq (1 + \varepsilon) \log \text{rad}(abc) + C_\varepsilon.$$

The Lean project defines:

```
def abcRadical (n : Nat) : Nat :=
  product p in n.primeFactors, p

def ABCConjecture : Prop :=
  forall epsilon : Real, 0 < epsilon ->
    exists C : Real, forall a b c : Nat,
      0 < a -> 0 < b -> 0 < c ->
      a + b = c -> PairwiseCoprimeABC a b c ->
      ABCLogInequalityAt epsilon C a b c
```

The source code uses Lean notation rather than the ASCII paraphrase displayed above.

2. THE VERIFIED IUT III CORRIDOR

2.1. **Poly-morphism compatibility.** Let

$$K_0 : A_0 \xrightarrow{\sim} C_0, \quad K_1 : A_1 \xrightarrow{\sim} C_1,$$

and let

$$H \subseteq \text{Iso}(A_0, A_1), \quad P \subseteq \text{Iso}(C_0, C_1)$$

be poly-isomorphisms. Define the sets of composite maps

$$K_1 H = \{K_1 \circ h : h \in H\}, \quad PK_0 = \{p \circ K_0 : p \in P\}.$$

Theorem 2.1 (Representative extraction). *If*

$$K_1 H = PK_0,$$

then every $h \in H$ has a compatible $p \in P$ satisfying

$$pK_0 = K_1 h.$$

Proof. The composite $K_1 h$ lies in $K_1 H = PK_0$. Membership in PK_0 supplies $p \in P$ with the desired equality. $\square$

This theorem is fully formalized. It replaces an earlier interface that simply assumed that the induced transport was allowed.

2.2. **Coherent pilot witness.** The structure `PolyCompatiblePilotData` adds:

- (i) a theta-pilot in A_0 ;
- (ii) a predicate identifying q -pilots in A_1 ;
- (iii) one horizontal representative that maps the theta-pilot to a q -pilot;
- (iv) a possible-output relation in C_1 containing the ordinary Kummer branch;
- (v) a monotone region-volume reading;
- (vi) the native q -pilot reading and the theta-hull reading.

Lean then proves the existence of $w \in C_1$ such that

$$w \in \text{possible}, \quad \text{volume}(\text{realize}(w)) = q_{\text{signed}}, \quad \text{realize}(w) \leq \theta_{\text{hull}}.$$

Consequently,

$$q_{\text{signed}} \leq \theta_{\text{signed}}.$$

With

$$q_{\text{signed}} = -Q, \quad Q > 0, \quad \theta_{\text{signed}} \leq C_{\Theta}Q,$$

Lean proves

$$C_{\Theta} \geq -1.$$

3. THE COMPLETE FORMAL DEPENDENCY THEOREM

The development defines a downstream interface:

Definition 3.1 (IUT IV reduction interface). For a real number C_{Θ} , an `IUTIVReduction` consists of a proof that

$$-1 \leq C_{\Theta} \implies \text{ABCConjecture}.$$

The full theorem is:

Theorem 3.2 (Formal IUT-*abc* dependency chain). *Given*

- (1) *PolyCompatiblePilotData*;
- (2) $Q > 0$;
- (3) *the native reading* $q_{\text{signed}} = -Q$;
- (4) *the theta upper estimate* $\theta_{\text{signed}} \leq C_{\Theta}Q$;
- (5) *IUTIVReduction* for C_{Θ} ;

the logarithmic abc conjecture holds.

Its Lean name is:

`abc_of_polycompatible_iut`

The proof is only a few lines because the difficult mathematics is concentrated in the two input structures.

4. WHY THIS IS NOT YET AN UNCONDITIONAL PROOF

A theorem with arguments

$$S + R \implies A$$

does not prove A unless closed terms of types S and R have been constructed.

Here:

$$S = \text{PolyCompatiblePilotData}, \quad R = \text{IUTIVReduction}, \quad A = \text{ABCConjecture}.$$

The first interface requires a first-principles encoding of:

- (i) Hodge theaters and their structured morphisms;
- (ii) the relevant Frobenioids and prime-strips;
- (iii) the exact poly-morphism composition law;
- (iv) Kummer isomorphisms on the pilot-forming splitting and global data;
- (v) the Ind1–Ind3 possible-image relation;
- (vi) the local-to-global fractional-ideal realization;
- (vii) procession-normalized log-volume and its compatibility.

The second interface requires formalizing:

- (i) the explicit local estimates of IUT IV;
- (ii) the general-position reduction;
- (iii) arithmetic heights, differentials, and conductors;
- (iv) the hyperbolic-curve height inequality;
- (v) its specialization to $(\mathbb{P}^1, \{0, 1, \infty\})$;
- (vi) the identification of the conductor with $\sum_{p|abc} \log p$.

Neither interface is currently a closed term in the supplied project.

5. KERNEL AUDIT

The project includes `AxiomAudit.lean`, which asks Lean to print the logical axioms used by:

- representative extraction;
- coherent witness construction;
- the ordered-volume bridge;
- `abc_of_polycompatible_iut`;
- the certificate theorem `FormalABCProofCertificate.abc`.

This audit is useful but has a precise interpretation. It shows that the *implications* use no hidden user-declared axiom. It does not show that the theorem arguments have inhabitants.

6. FORMALIZATION ARCHITECTURE

Lean module	Role
<code>ABCStatement.lean</code>	Exact logarithmic statement of <i>abc</i> and the radical.
<code>PolyCompatiblePilotWitness.lean</code>	Extracts a coherent representative and the native-volume possible output from equality of composite poly-morphisms.
<code>StructuredSourceBoundary.lean</code>	Ordered-volume bridge and $C_\Theta \geq -1$.
<code>IUTIVInterface.lean</code>	Explicit downstream arithmetic-geometric boundary.
<code>FullABCChain.lean</code>	Proves <code>ABCConjecture</code> from the two interfaces.
<code>AxiomAudit.lean</code>	Kernel dependency report for the abstract chain.

7. WHAT A GENUINE FINAL LEAN PROOF MUST LOOK LIKE

A genuinely closed theorem must have a declaration of the form

```
theorem abc_conjecture : ABCConjecture := by
  ...
```

with no parameters of types equivalent to `PolyCompatiblePilotData` or `IUTIVReduction`, and no imported axiom encoding either of them.

The proof term must construct:

`PolyCompatiblePilotData`

from the complete IUT I–III definitions, and

`IUTIVReduction`

from the complete IUT IV argument.

Only after both constructions kernel-check can `abc_conjecture` be advertised as a Lean-verified proof.

8. CURRENT MATHEMATICAL STATUS

The source does provide substantial support for the first interface. The IUT series defines poly-morphisms as collections of morphisms, treats full poly-isomorphisms as the collection of all isomorphisms, supplies functorial Kummer constructions, and states compatibility between horizontal and Kummer poly-isomorphisms. It also defines the theta volume using the union of Ind1–Ind3 possible images and the *q* volume using the native image.

The remaining issue is not the elementary extraction lemma; that is formalized. It is the complete type-correct instantiation of the lemma on the exact pilot-forming IUT data, together with the full downstream height theory.

Submission warning

The present project is suitable as a formalization blueprint and a verified conditional theorem. It is not suitable for submission under the title “A Lean-verified proof of the *abc* conjecture.” Such a title would assert that the two interface structures had been constructed, which the code itself shows has not yet occurred.

9. CONCLUSION

The requested “complete Lean proof from first principles” has not yet been obtained. What has been obtained is a precise statement of *abc*, a kernel-checkable proof of the full abstract corridor, an explicit theorem dependency graph, and a source-level target for continued formalization.

The next mathematically substantive task is not to add another wrapper theorem. It is to replace the field

`poly_compatibility`

by a theorem derived from concrete IUT types, and to replace

`IUTIVReduction`

by the formalized arithmetic-geometric proof. Until those replacements are made, honesty requires retaining the qualifier *conditional proof candidate*.

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